High-Efficiency Apodized Grating Coupler on Silicon-on-Insulator Platform with Aluminium Reflector

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Introduction

Silicon Photonics has emerged as a powerful platform for realising a high-speed optical communication system. So, Efficient coupling between the fibre and the on-chip waveguide remains a key challenge. Among various techniques, Grating Couplers are widely used due to their compatibility with planar fabrication and ease of integration. However, a significant limitation og Grating Coupler is the loss of Optical power due to the downward transition into the substrate, which reduces overall efficiency. In our work we have investigated the use of metal reflectors to redirect downward radiated power to enhance the coupling Efficiency.

Highlights

Results:

- Coupling Efficiency improved from 60% to 90%
- Silver Reflector produces the highest performance of all the materials tested
- Significant improvement in Substrate leakage.
- Simple modification to baseline design.
- Comparative analysis done for different metal reflectors.
- Strong TE Mode Confinement observed.
- Al is more preferable due to its CMOS compatibility

Key Advantages:

- .Achieving high efficiencies using reflector-assisted enhancement.
- Demonstrates material-dependent performance variations.

Design and Working

- The Grating Coupler is based on inverse 2D Design. It consists of an apodized grating structure on a silicon waveguide layer over the BOX Structure.
- Metallic Reflectors(Al,.Ag,Au) are introduced below BOX layer.
- The grating structure diffracts incident light from the fibre into the waveguide.
- Downward radiation losses into the substrate reduce efficiency in conventional designs
- A metallic reflector is used to reflect this lost power upward.
- This improves coupling efficiency and maintains strong TE mode confinement.

APODIZATION FUNCTION OF GRATING COUPLER:

- **Fill Factor Variation**:

$$\Lambda(x) = \frac{\lambda_c}{a+b(F_0-R(x-x_{start}))}$$

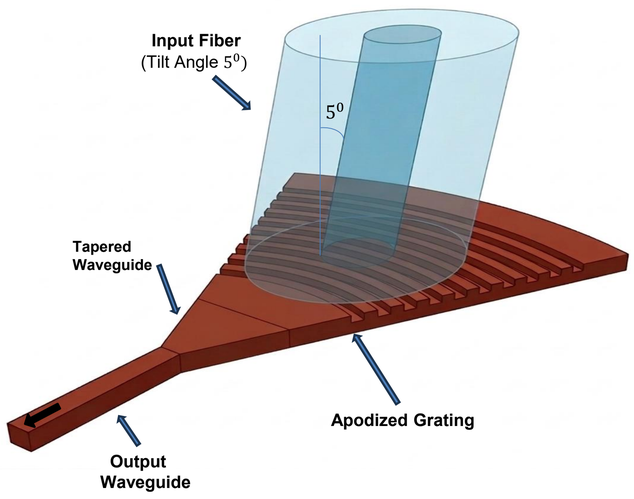
- **Grating Period Variation**:

$$F(x) = F_0 - R(x - x_{start})$$

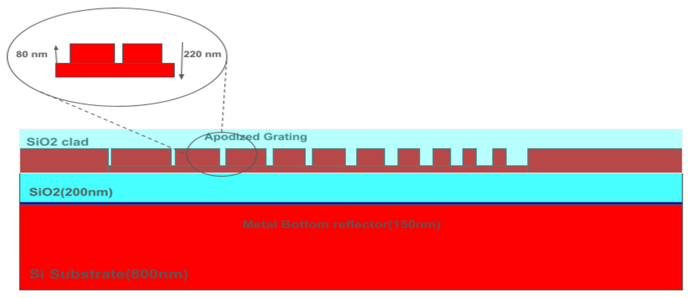
- **Combined Apodization Expression:**

$$\Lambda(x) = \frac{\lambda_c}{a+bF(x)}$$

Schematic of Grating Coupler

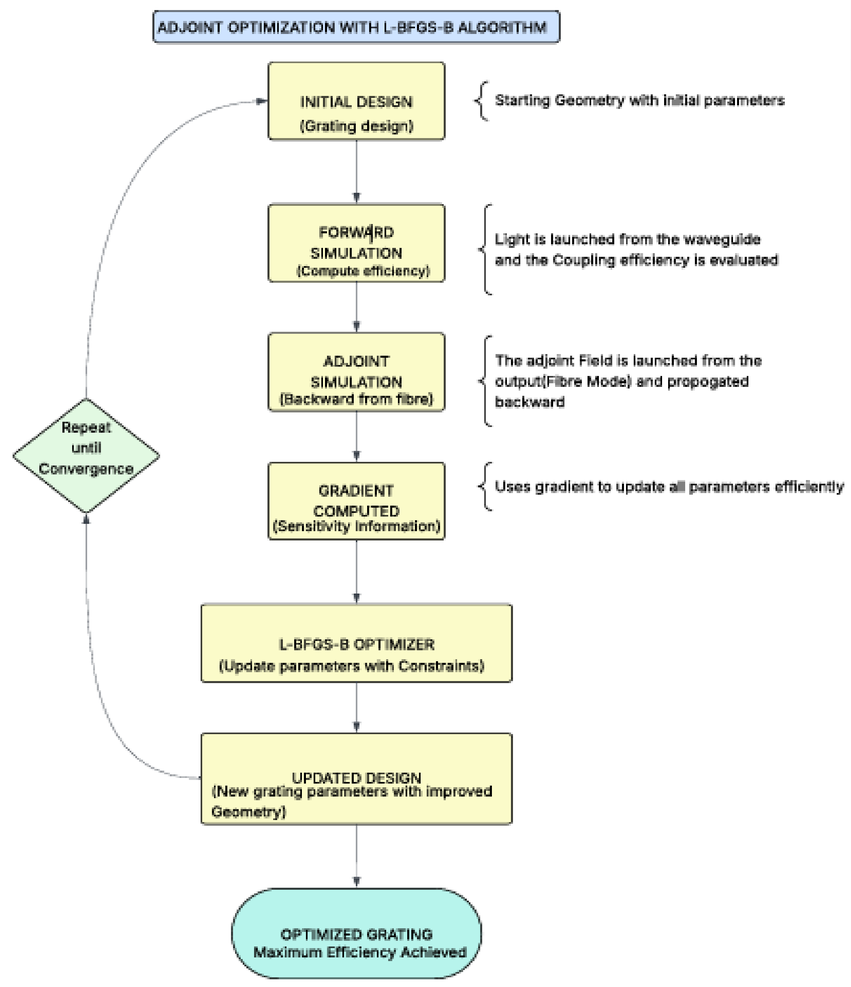


3D Structure of Apodized Grating Coupler



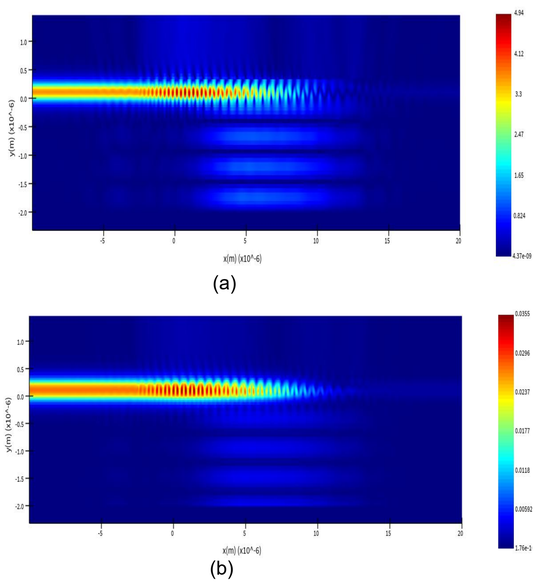
2D Cross-section of the side view of the grating coupler

Inverse Design Algorithm

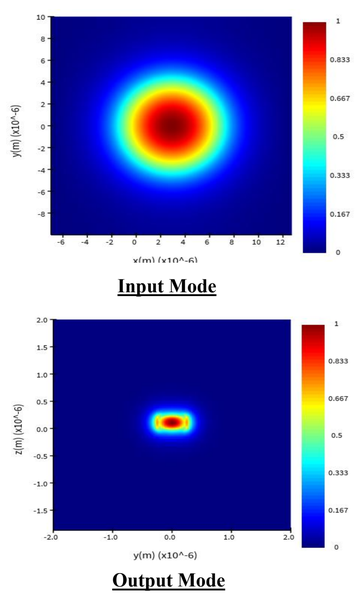


Result Analysis

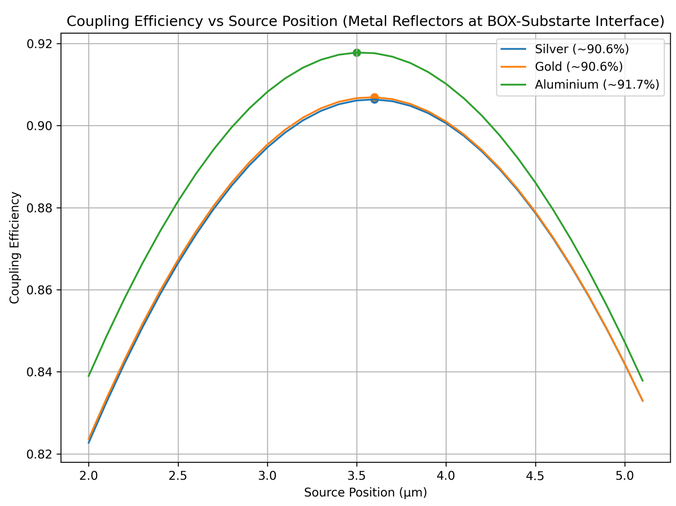
Field Profile for the (a) E and (b) H field

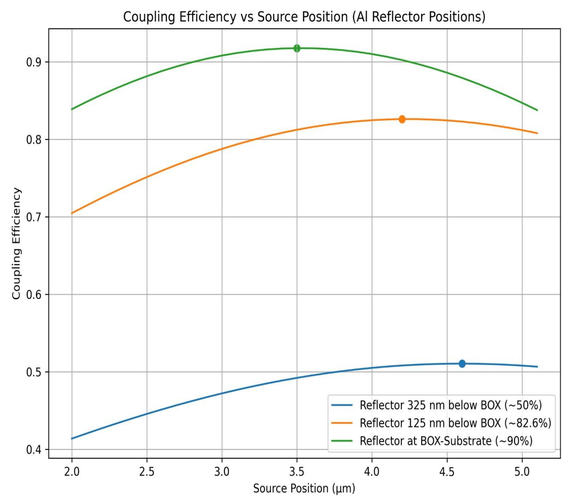


Fiber Input vs Waveguide Output



Coupling Efficiency with different Metal reflectors as a function of Source position





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